1. Cell Structure

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1. The Plasma Membrane

The plasma membrane is a thin barrier that separates the inside of the cell from its surroundings. It is built from a phospholipid bilayer, with hydrophilic heads facing the water and hydrophobic tails facing each other. Membrane proteins embedded in the bilayer act as channels, pumps, and receptors.

Diffusion is the movement of molecules from an area of high concentration to an area of low concentration. Osmosis is the diffusion of water across a selectively permeable membrane. Facilitated diffusion uses channel proteins to move molecules down their gradient without spending energy. Active transport moves molecules against their concentration gradient and requires energy from ATP. The sodium-potassium pump is a classic example of active transport in animal cells.

1. Organelles

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